

Fieldwork Data Lab - Pattern & Anomaly

I can check and present fieldwork data in a suitable graph, then interpret pattern, anomaly and reliability.

[Open the original interactive lesson](#)

Arrival · retrieve and reconnect

Standard

Retrieve one W11 pattern and identify one missing or unreliable value.

Starter · enquiry / tension

Enquiry

Which representation makes the fieldwork pattern clearest without distorting it?

QUESTION / HYPOTHESIS

A hypothesis is a claim the data can contradict.

MAP / DATA / METHOD

Right graph for the data: category per site → bar chart.

PATTERN + ANOMALY

Starter · enquiry / tension

Pattern: falling. Anomaly: site 5. Do not delete it — explain or re-sample it.

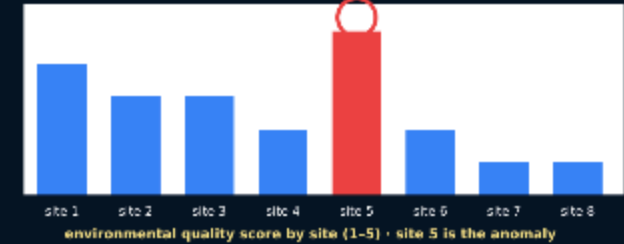
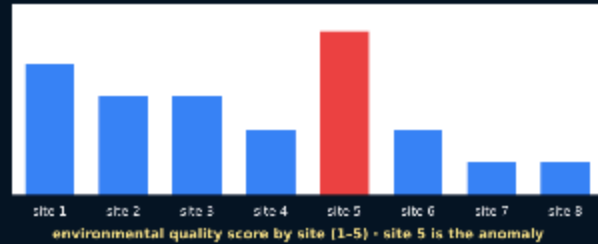
JUDGEMENT + LIMIT

Pattern, anomaly, reliability — all three in the write-up.

Starter · enquiry / tension

"Quality falls with distance from the park."

testable: one variable, one direction



Hypothesis

Bar chart of site scores

Anomaly circled

Starter · enquiry / tension

“Scores generally fall with distance, except site 5 (a new planting scheme).”

reliability: eight sites, one afternoon, one scorer — a pattern, not a proof

Judgement

I Do · model geographical reasoning

Model one reasoning step at a time.

Clean before graphing: preserve raw data, label units, retain blanks and flag impossible values.

Choose a bar chart for categories, a line graph for ordered change and a scatter graph for relationships between numerical variables.

Use a clear title, labelled axes, equal scale and visible constructed/source note.

Graph doctor · right graph, right verdict

I Do · model geographical reasoning

Data: a quality score (1–5) at eight named sites. Best graph?

- A. bar chart
- B. pie chart
- C. line graph

Which graph fits categories with a score each?

Site 5 scored 5 when the pattern predicts 2. What now?

- A. delete it
- B. keep it, flag it, explain or re-sample

I Do · model geographical reasoning

Anomalies are evidence too.

We Do · hinge judgement

Commit before reveal: Which graph best tests the relationship between distance and environmental-quality score?

- A. Pie chart
- B. Scatter graph
- C. Pictogram

I Do 2 · counter / qualification

Strengthen—not lengthen—the reasoning.

Describe a pattern with specific figures, then identify an anomaly before explaining.

Calculate and check summary values; do not let a formula hide a data-entry mistake.

Correlation is association, not proof that one variable caused the other.

We Do 2 · interactive reasoning lab

Data Bench · Graph Doctor

CONSTRUCTED CLASSROOM RESOURCE: This fictional composite is inspired by the broad Teesside setting. It is not official data, is not to scale and must not be used for navigation, safety or real-world planning.

Mini spreadsheet · calculate/check means

We Do 2 · interactive reasoning lab

Site	Counts	Your mean
Transit Gate	58, 63, 61	
Market Link	49, 46, 50	
Riverside Walk	26, 29, 27	

We Do 2 · interactive reasoning lab

- A. Footfall scatter
- B. EQS scatter
- C. Green scatter
- D. Noise scatter

Calculate/check first. Then inspect a scatter view, quote two values and retain anomaly and limitation.

Accessible data table

We Do 2 · interactive reasoning lab

Constructed fieldwork scatter graph

Independent · sustained evidence

Independent mission

I can check and present fieldwork data in a suitable graph, then interpret pattern, anomaly and reliability.

16 minutes. Approved access can change format, reading load or response mode; it cannot supply the geographical judgement.

Supported

Use the raw repeat counts to check three means, complete graph title/axis/unit prompts, then state a pattern, anomaly and reliability limit.

Independent · sustained evidence

Supported access: prepared table, formula cue and labelled axis frame; interpretation remains learner-owned.

Standard

Create a suitable graph and write two quantified pattern statements, one anomaly and one reliability or causation limit.

Standard scaffold: clean → calculate → choose → label → check → interpret.

Stretch

Independent · sustained evidence

Compare two graph choices, evaluate reliability and explain why the evidence cannot establish cause.

Stretch scaffold: test alternative graph, anomaly and correlation/causation limit.

Exit · evidence → response → Influence

VOICE: Give one quantified pattern, one anomaly and one reason the graph cannot prove cause.